

Class

B2IT / B2ITE

Object Oriented Programming (OOP)

(two hours)

Date: 7.5.2018

Name: _____

Reg. #: _____

10	8	8	6	4	14	

I. Choose the correct answers:

Q1. In a class definition, data or functions designated private are accessible

- (A) to any function in the program. (B) only to public members of the class.
(C) only if you know the password. (D) to member functions of that class.

Q2. A member function can always access the data

- (A) in any object of the class of which it is a member. (B) in the public part of its class.
(C) in the class of which it is a member. (D) in the object of which it is a member.

Q3. The dot operator (class member access operator) connects the two entities (left to right):

- (A) A class object and a member of that class (B) A class and a member of that class
(C) A class member and a class object (D) A class object and a class

Q4. Virtual functions allow you to

- (A) create an array of type pointer-to-base class that can hold pointers to derived classes.
(B) use the same function call to execute member functions of objects from different classes.
(C) group objects of different classes so they can all be accessed by the same function code.
(D) create functions that can never be accessed.

Q5. Inheritance is a way to

- (A) pass arguments to objects of classes.
(B) add features to existing classes without rewriting them.
(C) improve data hiding and encapsulation.
(D) make general classes into more specific classes.

Q6. Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access

- (A) private members of Derv. (B) public members of Derv.
(C) protected members of Derv. (D) public members of Base.

Q7. A friend function can be used to

- (A) allow access to an unrelated class.
(B) allow access to classes whose source code is unavailable.
(C) mediate arguments between classes.
(D) increase the versatility of an overloaded operator.

Q8. An abstract class is useful when

- (A) you want to defer the declaration of the class.
- (B) there are multiple paths from one derived class to another.
- (C) no objects should be instantiated from it.
- (D) no classes should be derived from it.

II. Fill in the blanks in each of the following:

(CHOOSE FIVE)

Q9. Member functions defined inside a class definition are _____ by default.

Q10. The only technical difference between structures and classes in C++ is that _____.

Q11. A constructor is executed automatically when an object is _____.

Q12. Deciding—after a program starts to execute—what function will be executed by a particular function call statement is called _____.

Q13. In the UML, inheritance is called _____.

Q14. To be accessed from a member function of the derived class, data or functions in the base class must be public or _____.

III. State whether each of the following is true or false. If false, explain why:
(CHOOSE FOUR)

Q15. In a class you can have more than one constructor with the same name.

Q16. Data items in a class must be private.

Q17. If a base class contains a member function `basefunc()`, and a derived class does not contain a function with this name, can an object of the derived class access `basefunc()`?

Q18. The user must always define the operation of the copy constructor.

Q19. A friend function can access a class's private data without being a member of the class.

IV. Direct questions:

Q20. What is the purpose of a class definition?

Q21. Explain the difference in operation between these two statements.

```
person p1(p0);
```

```
person p1 = p0;
```

Q22. If a base class and a derived class each include a member function with the same name, which member function will be called by an object of the derived class, assuming the scope-resolution operator is not used?

V. Write code of the following:

Q23. Assume that there is a class Derv that is derived from a base class Base . Write the declarator for a derived-class constructor that takes one argument and passes this argument along to the constructor in the base class.

Q24. Write the declaration for a pure virtual function called aragorn that returns no value and takes no arguments.

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VI. You have to choose ONLY ONE question 25 or 26.

- Q25. a) Illustrate the memory representation of static data members.
b) Write a mini-program that illustrates the virtual base classes.
c) Write the Rational class definition using the default parameter values constructor.
d) Write a simple program for illustrating the accessibility in the inheritance.

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Q26. a) Illustrate the virtual base classes by the UML.

b) Write a mini-program that illustrates multiple inheritance.

c) Write a class Counter definition code with the static count,

and including a constructor for increase the count and a destructor for decrease the count.

d) Write a simple program that shows how friend functions can act as a bridge between two classes.